

sender of messages encrypted with a symmetric group key.

Message counter synchronization is an essential part of enabling secure messaging between members of an operational group. Specifically, it protects against replay attacks, where an attacker replays older messages, which may result in unexpected behavior if accepted and processed by the receiver.

The recipient of a message encrypted with a group key can trust and process that message only if it has kept the last message counter received from a given sender using that key.

Underlying MCSP is a simple request / response protocol. When a multicast message with unknown counter is received, synchronization via MCSP begins by sending a synchronization request via unicast UDP to the multicast message originator's unicast IPv6 address. That originator then sends a synchronization response to the unsynchronized node via unicast UDP. After cryptographic verification, the formerly unsynchronized node is now synchronized with the originator's message counter and can trust the original and subsequent messages from the originator node.

4.18.1. Message Counter Synchronization Methods

There are two methods for synchronizing the message counter of a peer node, which are configurable per-group-key based on the [GroupKeySecurityPolicy field](#) of a given group key set (see [Section 11.2.5.4, "GroupKeySetStruct Type"](#)).

4.18.1.1. Trust-first policy

The first authenticated message counter from an unsynchronized peer is trusted, and its message counter is used to configure message-counter-based replay protection on future messages from that node. All control messages (any message with [C Flag](#) set) use the control message counter and SHALL use Trust-first for synchronization. Note that MCSP is not used for Trust-first synchronization.

This policy provides lower latency for less security-sensitive applications such as lighting.

WARNING

Trust-first synchronization is susceptible to accepting a replayed message after a Node has been rebooted.

4.18.1.2. Cache-and-sync policy

The message that triggers message counter synchronization is stored, a [message counter synchronization exchange](#) is initiated, and only when the synchronization is completed is the original message processed. Cache-and-sync provides replay protection even in the case where a Node has been rebooted, at the expense of higher latency.

Support for the cache-and-sync policy and [MCSP](#) is optional. A node indicates its ability to support this feature via the [Group Key Management](#) cluster [FeatureMap](#).

WARNING

Large groups may cause significant message synchronization burden on the sender and message queueing pressure or latency on the receiver when message counter synchronization state is out of sync, such as after a brown-out, on